

Based on your CV, an interviewer will likely ask about **your experience, AI concepts, Python, LLMs, RAG, LangChain, projects, and problem-solving**. Since this is for an **AI Engineer** role, here are realistic interview questions with strong sample answers.

1. Tell me about yourself.

Answer

My name is Muhammad Shafiullah Khan. I completed my BS in Computer Science from the University of Mianwali. Currently, I am working as an AI Engineer Trainee at Crewlogix Technologies, where I build AI applications using Python, LangChain, LangGraph, LangFlow, and Large Language Models.

I have hands-on experience developing AI HR systems, AI CV Builders, RAG-based chatbots, and workflow automation. My interests are Generative AI, Agentic AI, Retrieval-Augmented Generation, and scalable AI applications. I enjoy solving real-world business problems using LLMs.

2. Explain your AI HR Assistant project.

Answer

Explain using this format:

Problem

"We wanted to automate the HR recruitment process."

Features

- Resume upload
- Resume parsing
- ATS score
- Candidate ranking
- Job matching
- Interview question generation
- AI-generated offer letters

Tech Stack

- Python
- FastAPI
- LangChain
- LangGraph
- ChromaDB
- Groq/OpenAI LLM
- SQLAlchemy

Workflow

Resume



Extract Text



Chunking



Embedding



Vector Database



LLM



Response

3. What is RAG?

Answer

RAG stands for Retrieval-Augmented Generation.

Instead of answering only from the LLM's knowledge, it first retrieves relevant information from documents and then provides an answer using that retrieved context.

Workflow

Question

↓

Embedding

↓

Similarity Search

↓

Top-k Documents

↓

LLM

↓

Answer

Benefits

- Less hallucination
- Uses company documents
- More accurate
- Up-to-date information

4. Explain embeddings.

Answer

Embeddings convert text into numerical vectors.

Similar meanings have vectors that are close together, allowing semantic search.

Example

"I love AI"

↓

[0.23, 0.76, 0.12...]

5. Why do we need vector databases?

Answer

Normal SQL searches exact text.

Vector databases search by meaning.

Examples

- ChromaDB
 - FAISS
 - Pinecone
 - Weaviate
 - Milvus
-

6. Difference between SQL database and Vector Database?

Answer

SQL

- Structured data
- Exact search
- Tables

Vector DB

- Semantic search
 - Embeddings
 - AI applications
-

7. What is LangChain?

Answer

LangChain is a framework for building LLM applications.

It provides

- Prompt Templates
 - Chains
 - Agents
 - Memory
 - Tools
 - Retrieval
 - Output Parsers
-

8. Why LangGraph instead of LangChain?

Answer

LangChain works well for sequential workflows.

LangGraph is designed for complex agent workflows with branching, looping, conditional routing, checkpoints, and persistent state.

9. What is LangFlow?

Answer

LangFlow is a visual interface for building LangChain pipelines using drag-and-drop components.

Useful for rapid prototyping.

10. Explain Agentic AI.

Answer

Agentic AI can

- Think
- Plan
- Use tools

- Make decisions
- Execute multiple steps

Example

HR Assistant

User asks

"Find the best candidate."

Agent

Reads resumes

↓

Scores resumes

↓

Ranks candidates

↓

Generates report

11. What is Prompt Engineering?

Answer

Prompt engineering is designing prompts that guide the LLM to produce accurate, relevant, and consistent responses.

12. What is Chain of Thought?

Answer

The model reasons step by step before producing the final answer.

Useful for

- Math
 - Logic
 - Coding
-

13. What is Tree of Thought?

Answer

Instead of following one reasoning path, the model explores multiple reasoning paths and chooses the best one.

14. Difference between Chain of Thought and Tree of Thought?

Chain of Thought

One reasoning path.

Tree of Thought

Multiple reasoning paths.

15. What LLMs have you used?

Example Answer

I have worked with

- Llama 3
 - TinyLlama
 - Groq-hosted models
 - OpenAI GPT APIs (if applicable)
-

16. How do you reduce hallucinations?

Answer

- Use RAG
 - Better prompts
 - Low temperature
 - Verify retrieved context
 - Guardrails
 - System prompts
-

17. Explain chunking.

Answer

Large documents are divided into smaller chunks before creating embeddings.

Why?

Because LLM context windows are limited.

18. Which embedding model have you used?

Example

all-MiniLM-L6-v2

19. Explain your AI CV Builder.

Answer

Workflow

User Information

↓

Job Description



Prompt Engineering



LLM



ATS Friendly Resume



Download PDF

20. Explain FastAPI.

Answer

FastAPI is a high-performance Python framework for building REST APIs.

Advantages

- Fast
 - Async support
 - Automatic Swagger documentation
 - Easy validation using Pydantic
-

21. Explain your API structure.

Example

POST /upload-resume

POST /rank-candidates

POST /generate-cv

GET /candidate

POST /chat

22. Explain your Student Guide System.

Answer

Students ask questions.

↓

RAG retrieves university policies.

↓

LLM generates answers.

↓

Memory stores conversation.

23. Explain your Bus Tracking project.

Possible Questions

Why AI?

How does tracking work?

How would you improve it?

24. Have you deployed applications?

Answer

Yes.

I have deployed frontend applications using Vercel and have experience building backend APIs with FastAPI. I also understand Docker for containerizing AI applications.

25. Explain Docker.

Answer

Docker packages an application along with its dependencies so it runs consistently across development, testing, and production environments.

26. What challenges did you face?

Good Answer

Some challenges included:

- Hallucinations
 - Slow retrieval
 - Chunking strategy
 - API rate limits
 - Prompt optimization
 - Dependency conflicts
 - Memory management
-

27. How do you improve RAG accuracy?

Answer

- Better chunking
 - Better embeddings
 - Reranking
 - Metadata filtering
 - Query rewriting
 - Hybrid search
 - High-quality documents
-

28. Difference between Fine-tuning and RAG?

RAG

- External knowledge
- No model retraining
- Easy updates

Fine-tuning

- Retrains model weights
 - More expensive
 - Best for changing behavior or specialized tasks
-

29. Explain your Git workflow.

Answer

```
git clone
```

```
git checkout -b feature
```

```
git add
```

```
git commit
```

```
git push
```

```
Create Pull Request
```

30. Why should we hire you?

Answer

I have practical experience building AI applications rather than only studying AI concepts. I have worked on LLMs, RAG pipelines, LangChain, LangGraph, FastAPI, Python, and React to build end-to-end AI solutions. I enjoy learning new technologies quickly, solving real-world problems, and collaborating with teams. I believe I can contribute from day one while continuing to grow as an AI Engineer.

Coding Questions You May Be Asked

- Reverse a string in Python.
- Check if a string is a palindrome.
- Find duplicate elements in a list.
- Two Sum problem.
- Missing Number problem.
- Merge two sorted arrays.
- Reverse a linked list.
- Explain list vs tuple.
- Difference between `==` and `is`.
- Explain decorators.
- Explain generators.
- Explain `*args` and `**kwargs`.
- Explain multithreading vs multiprocessing.
- Write a REST API in FastAPI.

Scenario-Based AI Questions

1. A user asks a question that is not in your RAG documents. How will your system respond?
2. Your chatbot gives the wrong answer. How will you debug it?
3. How would you improve retrieval quality in a RAG pipeline?
4. Which embedding model would you choose for a multilingual chatbot, and why?
5. How would you design a multi-agent system for HR recruitment?
6. How do you secure LLM APIs and protect sensitive data?
7. When would you choose fine-tuning over RAG?
8. How would you optimize the latency of an AI application?

Preparing clear answers to these questions—and being able to explain the architecture and trade-offs in your own projects—will put you in a strong position for an AI Engineer interview.